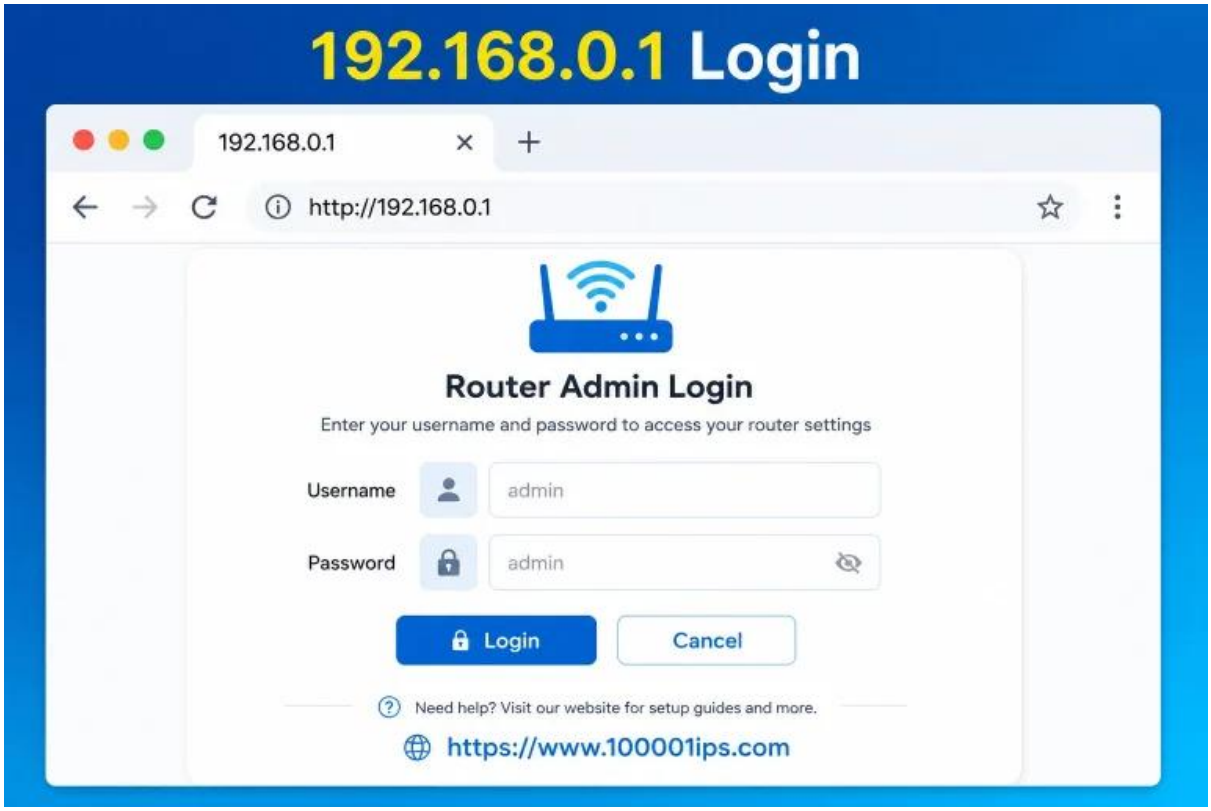
1.

Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.



2.

Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Wi-Fi Network Details

Network Detail	Example Value
IP Address	192.168.1.105
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

3.

Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Change DHCP to Static IP

Windows laptop – simple steps:

1. Connect your laptop to your **Wi-Fi**.
2. Open **Settings → Network & Internet → Wi-Fi**.
3. Open your connected Wi-Fi network and select **IP assignment → Edit**.
4. Change **Automatic (DHCP)** to **Manual**.
5. Turn on **IPv4**.
6. Enter example values:
    - **IP Address:** 192.168.1.50
    - **Subnet Mask:** 255.255.255.0
    - **Gateway:** 192.168.1.1
    - **Preferred DNS:** 8.8.8.8
7. Click **Save**.
8. Open a browser and check whether you can access the internet.
9. After testing, go back to **IP assignment → Edit**.
10. Change **Manual** back to **Automatic (DHCP)** and save.

4.

In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

DHCP Range

Example:

- **DHCP Range:** 192.168.1.100 – 192.168.1.200
- **Router/Gateway:** 192.168.1.1

What happens if two devices get the same IP?
If two devices have the same IP address, an **IP conflict** occurs. Both devices may have problems connecting to the internet or communicating with other devices on the network.